

Benjamin Karachun

Biotechnology & Food Engineer | Full-Stack Developer Israel | 054-612-3730 | kbenjamin@campus.technion.ac.il | [LinkedIn](#) | [GitHub](#)

PROFILE

Technion Biotechnology student (Sem 6) bridging the gap between Life Sciences and Software Engineering. Unique combination of **wet-lab proficiency** and **full-stack development** (React, Python, Firebase). Passionate about Computational Biology and fighting antibiotic resistance (*S. aureus* focus). Proven track record of building tools that solve real-world academic and research problems.

EDUCATION

Technion - Israel Institute of Technology | Haifa, Israel **B.Sc. in Biotechnology and Food Engineering** | Expected 2027

- Specialization:** Computational Biotechnology & Food Engineering and Health.

Bagrut | Mofet Excellence Program

- 5 Units in:** Chemistry, Physics, Engineering Sciences, Math, English.

TECHNICAL PROJECTS

BiotechFood Prep Platform | Founder & Developer | bfe.technionprep.com

- Architected and deployed a full-stack educational platform serving Technion students.
- Built with **React** (Frontend) and **Firebase** (Backend, Hosting, Google OAuth).
- Created interactive study tools and resource repositories, directly improving student study efficiency.

Shoresh Nutrition App | Founder & Developer | shoresh.streamlit.app

- Developed a data-driven nutrition calculator and label creator leveraging the Israeli Ministry of Health database.
- Engineered a Python ETL pipeline to clean and structure government data into SQLite.
- Built a **Streamlit** interface with Hebrew localization and dynamic unit conversion.

MathPaste | Founder & Developer (Open Source)

- Created a Chrome Extension to solve the "scientific input" problem on the web.
- Implemented a WYSIWYG editor that injects LaTeX/MathJax directly into any text field.
- Tech:** JavaScript, Chrome APIs, DOM manipulation.

AlgoPrep | Founder & Lead Educator | algoprep.com | TikTok: [@algo_prep](#)

- Founded a coding education brand with thousands of followers on TikTok.
- Built a marketing funnel converting social media engagement into a client base of **dozens of private students**.
- Created content focusing on **Algorithm Analysis** and **Developer Efficiency** (VS Code workflows, shortcuts, and optimization).

SKILLS

AI & LLM Engineering:

- Early Adopter:** Experience with LLMs since GPT-3 (Davinci). Deep understanding of prompt engineering architectures.
- Prompt Architecture:** Designed sophisticated, reusable prompt libraries for educational content generation. Solved complex technical challenges in AI output (RTL/LTR directionality, MathJax 3 integration, strict JSON schema validation).
- Frameworks:** **Antigravity**, **OpenClaw** (Advanced Agentic Workflows), LangChain concepts.
- Integration:** Building apps with OpenAI/Gemini/Claude APIs. Automating Linux environments and research workflows using AI agents.

Software & Data:

- Languages:** Python (NumPy, Pandas), JavaScript/TypeScript, SQL.
- Web:** React, HTML5/CSS3, Firebase, Streamlit, Chrome Extensions.
- Tools:** Git, Linux/Bash, Docker, SSH.

Laboratory (Wet Lab):

- Molecular:** PCR, Gel Electrophoresis.
- Microbiology:** Aseptic techniques, Pour plate/Spread plate, Drigalski spatula method.
- Instrumentation:** Stomacher, 96-well Plate Readers, Spectrophotometry.

Languages:

- Hebrew (Native)
- English (Fluent)
- Russian (Fluent)

EXPERIENCE

Python Tutor | Technion Student Community

- Mentored undergraduate students in the Technion's Python programming course
- Architected and deployed **Python TechnionPrep** (python.technionprep.com), a dedicated educational platform designed to centralize study materials, streamline the learning process enhance student outcomes.

Digital Transformation Consultant | Hanan Shemesh Accounting

- Advised on and implemented technological modernizations for an accounting firm.
- Integrated **WhatsApp Business API** to automate client communication and request handling.
- Led the digitalization process of legacy paper records, improving data accessibility and security.

Web Maintenance & Development | Aviator.co.il

- Managed and maintained WordPress infrastructure for a commercial client.
- Handled updates, asset management, and troubleshooting.